CSE 126N PROJECT – FALL 2023

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This document presents CSE 126N Project , The Company Management System (CMS) project is an essential educational endeavor developed for the Computer & Communication Program at Alexandria National University. Its development is a direct response to the need for a straightforward and efficient tool for managing employee records in a corporate setting. The CMS is designed to be user-friendly, ensuring that users can easily navigate through various functionalities without the need for advanced technical knowledge.

The system's primary goal is to maintain an organized database of employee information. This includes vital details such as identification numbers, full names, salaries, dates of birth, addresses, contact numbers, employment start dates, and email addresses. A distinctive feature of the CMS is its ability to enforce unique identifiers for each employee, preventing duplicate entries and ensuring the accuracy of the data.

Functionality is at the heart of the CMS, with the system supporting a wide range of user commands. These include loading employee data from a file, adding new entries, updating existing ones, deleting records, searching for specific employees by name, and sorting the records based on various criteria such as name, salary, or date of birth.

The system is particularly adept at handling large volumes of data, This is crucial for organizations that need to migrate their existing employee data into the CMS. The search function is designed to be forgiving, allowing partial name matches to return relevant results, thus enhancing the user experience.

When modifications are made to the database, such as adding or updating records, the CMS automatically assigns the current date to the date of enrollment field, reducing manual entry errors. Moreover, the CMS is equipped with a sorting function that organizes the data in a clear and understandable manner, which is essential for reporting and decision-making processes.

An essential aspect of the CMS is its ability to save the updated employee database back to the original file, ensuring that all changes are preserved for future sessions. A warning system is also in place to alert users if they attempt to exit the program without saving, thus preventing accidental data loss.

Upon launching, the CMS presents a straightforward menu that allows the user to select from the various available options, ensuring a seamless interaction with the system. The default action upon starting the program is to load the existing employee data, thereby preparing the system for immediate use.

In conclusion, the Company Management System is a vital tool tailored to enhance the management of employee information within a company. Its simple yet powerful features are designed to meet the educational goals of the course and to provide students with practical experience in the development and use of management systems.

The CMS project consists of several source code that implement the core functionality and features of the system. These include:

main: The main function is responsible for initializing the system, loading the employee data from a file, and calling the menu function. The menu function displays the available options for the user and calls the corresponding functions based on the user's input.

add: **The add function and the helper functions for validating and formatting the employee data. The add function allows the user to add a new employee record to the system, after checking the validity of the input data and assigning a unique identifier to the employee.**

delete: **delete function and the helper function for finding an employee by ID. The delete function allows the user to delete an existing employee record from the system, after asking for the employee ID and confirming the deletion.**

modify: **The modify function and the helper functions for finding an employee by ID and updating the employee data. The modify function allows the user to modify an existing employee record in the system, after asking for the employee ID and the field to be modified.**

search: **The search function and the helper function for finding an employee by name. The search function allows the user to search for an employee record in the system by entering a partial or full name, and displays the matching results.**

print: **The print function and the helper functions for sorting and printing the employee data. The print function allows the user to print the employee records in the system in a sorted order, based on the user's choice of sorting criteria.**

save: save function and the helper function for writing the employee data to a file. The save function allows the user to save the changes made to the employee records in the system to the original file, ensuring that the data is preserved for future sessions.

quit: quit function and the helper function for exiting the system. The quit function allows the user to exit the system, after asking if the user wants to save the changes or not, and displaying a farewell message.

The data structures include:

struct date: A structure that represents a date, consisting of three integer fields: day, month, and year.

struct employee: A structure that represents an employee record, consisting of several fields: id (integer), name (string), salary (float), birth (date), address (string), mobile (long), enrollment (date), and email (string).

CONTENTS

CHAPTER 1. Sample Runs.

1. Sample Run for “Add function”

File Before :

data.txt

```
123, Steven Thomas, 20000, 10-6-1994, 26 Elhoreya Street, 01234567899, 15-9-2022, sthomas@gmail.com
```

Input :

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  1
Enter employee ID : 122
Enter employee full name : mohammed
Enter employee salary : 2500
Enter employee  birth date (dd/mm/yyyy) : 12/12/2004
Enter employee address : 12 alexndria
Enter employee mobile number : 01111111111
Enter employee email: mm@gmail.com
Added Successfully
```

File after save :

data.txt

```
123, Steven Thomas, 20000, 10-6-1994, 26 Elhoreya Street, 01234567899, 15-9-2022, sthomas@gmail.com
122, mohammed, 2500, 12-12-2004, 12 alexndria, 01111111111, 1-1-2024, mm@gmail.com
```

CHAPTER 1. Sample Runs.

2. Sample Run for "Delete function"

File Before :

data.txt

```
123, Steven Thomas, 20000, 10-6-1994, 26 Elhoreya Street, 01234567899, 15-9-2022, sthomas@gmail.com
122, mohammed, 2500, 12-12-2004, 12 alexndria, 01111111111, 1-1-2024, mm@gmail.com
```

Input :

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  2
Enter the Id of the employee you want to delete : 122
delete Successfully
```

File after save :

data.txt

```
123, Steven Thomas, 20000, 10-6-1994, 26 Elhoreya Street, 01234567899, 15-9-2022, sthomas@gmail.com
```

CHAPTER 1. Sample Runs.

3. Sample Run for "MODIFY function"

File Before :

```
data.txt
123, Steven Thomas, 20000, 10-6-1994, 26 Elhoreya Street, 01234567899, 15-9-2022, sthomas@gmail.com
```

Input :

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  3
Enter the Id of the employee you want to modify : 123
What you want to moodify
1-Name
2-salary
3-mobile
4-address
5-email
 Enter the number of data you want to modify : 2
Enter the new data
15000
modifded Successfully
```

File after save :

```
data.txt
123, Steven Thomas, 15000, 10-6-1994, 26 Elhoreya Street, 01234567899, 15-9-2022, sthomas@gmail.com
```

CHAPTER 1. Sample Runs.

4. Sample Run for “MODIFY function”

File Before :

data.txt — تم التعديل

```
123, Steven Thomas, 15000, 10-6-1994, 26 Elhoreya Street, 01234567899, 15-9-2022, sthomas@gmail.com
13, Mohammed salah, 20000, 10-6-2004, 26 Street, 01111111111, 15-9-2002, mm@gmail.com
10, Mahmoud, 22000, 10-9-2001, 26 Street, 01111111881, 15-9-2021, mm@gmail.com
```

Input & Output:

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  4
Enter employee name: M
13, Mohammed salah, 20000, 10- 6-2004, 26 Street, 01111111111, 15-9-2002, mm@gmail.com
10, Mahmoud, 22000, 10- 9-2001, 26 Street, 01111111881, 15-9-2021, mm@gmail.com
```

CHAPTER 1. Sample Runs.

5. Sample Run for “Print function”

File Before :

تم التعديل — data.txt

```
123, Steven Thomas, 15000, 10-6-1994, 26 Elhoreya Street, 01234567899, 15-9-2022, sthomas@gmail.com
13, Mohammed salah, 20000, 10-6-2004, 26 Street, 01111111111, 15-9-2002, mm@gmail.com
10, Ahmed, 22000, 10-9-2001, 26 Street, 01111111881, 15-9-2021, mm@gmail.com
```

Input & Output:

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  5
What type of sort You want ?
1-Name
2-Salary
3-Date of Birth : 2
10, Ahmed, 22000, 10- 9-2001, 26 Street, 01111111881, 15-9-2021, mm@gmail.com
13, Mohammed salah, 20000, 10- 6-2004, 26 Street, 01111111111, 15-9-2002, mm@gmail.com
123, Steven Thomas, 15000, 10- 6-1994, 26 Elhoreya Street, 01234567899, 15-9-2022, sthomas@gmail.com
```

CHAPTER 1. Sample Runs.

6. Sample Run for “ID check”

Input & Output:

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  1
Enter employee ID : f
You did not enter a valid number. Please try again.
Enter employee ID : 123
You entered a duplicated ID.
Enter employee ID : 122
Enter employee full name :
```

7. Sample Run for “Name check”

Input & Output:

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  1
Enter employee ID : 124
Enter employee full name : mm12@ salah
the eneterd data is not a text , repate it again
Enter employee full name : mohammed salah
Enter employee salary :
```

CHAPTER 1. Sample Runs.

6. Sample Run for "Salary check"

Input & Output:

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  1
Enter employee ID : 124
Enter employee full name : mm
Enter employee salary : q1213
You did not enter a valid number. Please try again.
Enter employee salary : 12300
Enter employee  birth date (dd/mm/yyyy) :
```

7. Sample Run for "Birth check"

Input & Output:

Format must be dd/mm/yyyy

Employee must be +18 years old

The month must have the number of day you entered

Counterexample : February can't have 30 day

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  1
Enter employee ID : 124
Enter employee full name : mm
Enter employee salary : 12500
Enter employee  birth date (dd/mm/yyyy) : 121222
Wrong year entered, add the date again
Enter employee  birth date (dd/mm/yyyy) : 12/12/1800
Wrong year entered, add the date again
Enter employee  birth date (dd/mm/yyyy) : 31/2/2001
Wrong day entered, add the date again
Enter employee  birth date (dd/mm/yyyy) : 22/2/2010
Employee must be at least 18 years old
Enter employee  birth date (dd/mm/yyyy) : 12/12/2004
Enter employee address :
```

CHAPTER 1. Sample Runs.

8. Sample Run for "Mobile check"

Input & Output:

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  1
Enter employee ID : 121
Enter employee full name : mm
Enter employee salary : 12500
Enter employee  birth date (dd/mm/yyyy) : 12/12/2004
Enter employee address : 12 sidi
Enter employee mobile number : qqoij13121
You did not enter a valid number. Please try again.
Enter employee mobile number : 011111
wrong mobile entered , add the mobile again
Enter employee mobile number : 01111111111
Enter employee email:
```

9. Sample Run for "Email check"

Input & Output:

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  1
Enter employee ID : 121
Enter employee full name : mm
Enter employee salary : 12500
Enter employee  birth date (dd/mm/yyyy) : 12/12/2004
Enter employee address : 12 sidi
Enter employee mobile number : 01111111111
Enter employee email: 11@gg.cc
The email's name part should contain at least one letter.
Enter employee email: q1@s%.cc
The email's domain part should contain only letters, numbers, and '-'.
Enter employee email: q1@gg.o1
The email's TLD part should contain only letters.
Enter employee email: mm@mm.com
```

CHAPTER 1. Sample Runs.

9. Sample Run for "Delete employee with wrong id "

Input & Output:

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  2
Enter the Id of the employee you want to delete : 2231
You entered a wrong ID. Please try again.
```

CHAPTER 2. User Manual.

After run the Program a function menu will open chose the the number of the function you want :

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :
```

Case 1 :

Add function

When you chose add , program will ask you for the employee information :

Id, Name, Salary, Date of birth, Address, Mobile, Email.

Notes :

Id should not be duplicated.

Name should contain alphabet and space only.

The salary should be valid .

Birth date should be in (dd/mm/yyyy) format.

The mobile number should be containing the 0 and other 10 digit .

Email should be valid .

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  1
Enter employee ID : 1234
Enter employee full name : mohammed salah
Enter employee salary : 12500
Enter employee  birth date (dd/mm/yyyy) : 12/12/2004
Enter employee address : 12 sidi
Enter employee mobile number : 01111111111
Enter employee email: mm@gmail.com
Added Successfully
```

CHAPTER 2. User Manual.

Case 2 :
Delete function.

When you chose Delete , program will ask you for the employee information : Id.

Notes :
Id should be In the employee otherwise program will ask for the ID again.

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  2
Enter the Id of the employee you want to delete : 123
delete Successfully
```

Case 3 :
Modify function.

When you chose Modify , program will ask you for the employee information : ID, Date you want to modify.

Notes :
Id should be In the employee otherwise program will ask for the ID again.

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  3
Enter the Id of the employee you want to modify : 123
What you want to moodify
1-Name
2-salary
3-mobile
4-address
5-email
 Enter the number of data you want to modify : 1
Enter the new data
mohammed
modifded Successfully
```

CHAPTER 2. **User Manual.**

Case 4 :

Search function.

When you chose Search , program will ask you for the employee information : (Part / Full) name.

Program will display all employee with the name you entered

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  4
Enter employee name: M
13, Mohammed salah, 20000, 10- 6-2004, 26 Street, 01111111111, 15-9-2002, mm@gmail.com
```

Case 5 :

Print By sort function.

When you chose Print , program will ask you for : Data You want to sort based on it.

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  5
What type of sort You want ?
1-Name
2-Salary
3-Date of Birth
Enter sort type : 1
10, Ahmed, 22000, 10- 9-2001, 26 Street, 01111111881, 15-9-2021, mm@gmail.com
13, Mohammed salah, 20000, 10- 6-2004, 26 Street, 01111111111, 15-9-2002, mm@gmail.com
123, Steven Thomas, 15000, 10- 6-1994, 26 Elhoreya Street, 01234567899, 15-9-2022, sthomas@gmail.com
```

CHAPTER 2. User Manual.

Case 6 :

Save function.

When you chose Save , Program will save the new employee written manually in file .

Note : Program will exit if employee Format is wrong.

Case 7 :

Quit function.

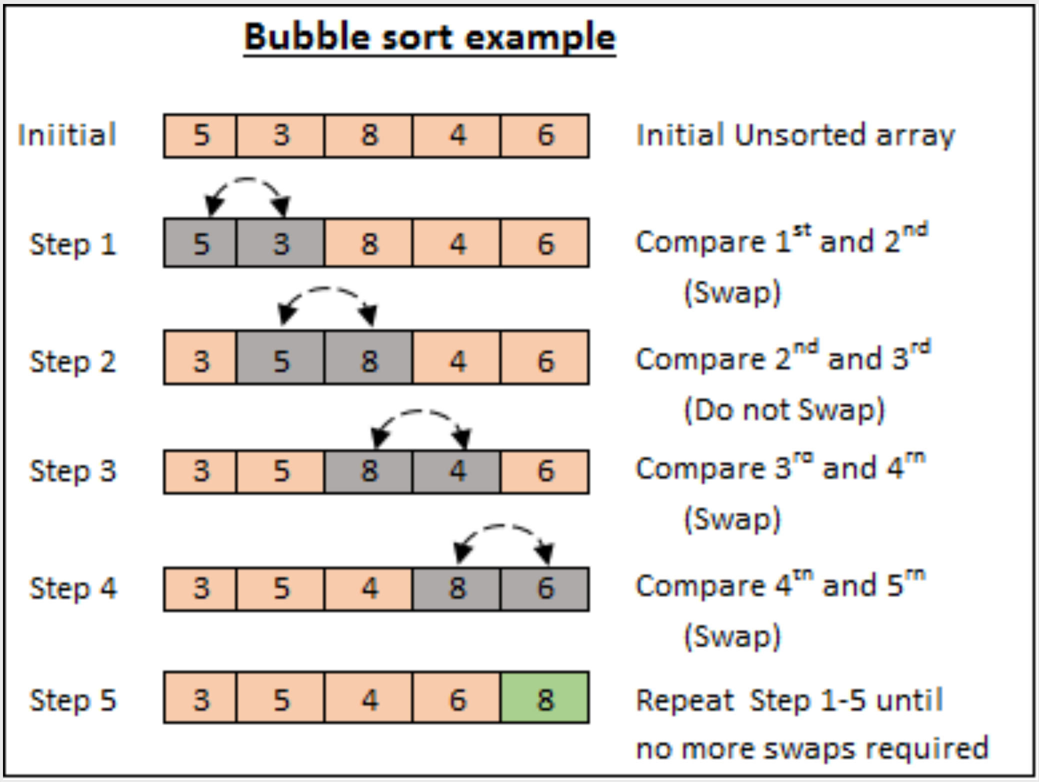
When you chose Quit , Program will ask you for that you want to save the changed in data you do in this run or not .

```
Choose the function
 1-ADD
 2-DELETE
 3-MODIFY
 4-SEARCH
 5-PRINT
 6-SAVE
 7-QUIT
 By enter the number of the function :  7
Are you want to exit without saving the data?
Enter :
1 for not saving
2 For saving
any number else for show menu again :
```

CHAPTER 3. Algorithms.

1. sort algorithms

All sort algorithm used in the program use bubble sort algorithm.



CHAPTER 3. Algorithms.

1.A Name sort algorithms

Algorithm.3.1.A .

```
void sorted_name(employee information[],int a){
employee sort[1000];
employee temp;
int i,j;
for (i=0;i<a;i++){

sort[i]=information[i];
}
for(i=0;i<a-1;i++){
for (j=0;j<a-1;j++){
if(strcmp(sort[j].name,sort[j+1].name)>0){
temp=sort[j];
sort[j]=sort[j+1];
sort[j+1]=temp;
}


}
}
for (i=0;i<a;i++){
printf("%d, %s, %d, %d-%d-%d, %s, 0%ld, %d-%d-%d, %s\n",sort[i].id,
sort[i].name,
sort[i].salary, sort[i].birth.day,
sort[i].birth.month, sort[i].birth.year,
sort[i].address, sort[i].mobile,
sort[i].enrollment.day,
sort[i].enrollment.month,
sort[i].enrollment.year, sort[i].email);
}
}
```

CHAPTER 3. Algorithms.

1.B Salary sort algorithms

Algorithm.3.1.B .

```
void sorted_salary(employee information[],int a){
employee sort[1000];
employee temp;
int i,j;
for (i=0;i<a;i++){

sort[i]=information[i];
}
for(i=0;i<a-1;i++){
for (j=0;j<a-1;j++){
if(sort[j].salary<sort[j+1].salary){
temp=sort[j];
sort[j]=sort[j+1];
sort[j+1]=temp;
}

}
}
for (i=0;i<a;i++){
printf("%d, %s, %d, %d-%d-%d, %s, 0%ld, %d-%d-%d, %s\n",sort[i].id,
sort[i].name,
sort[i].salary, sort[i].birth.day,
sort[i].birth.month, sort[i].birth.year,
sort[i].address, sort[i].mobile,
sort[i].enrollment.day,
sort[i].enrollment.month,
sort[i].enrollment.year, sort[i].email);
}
}
```

CHAPTER 3. Algorithms.

1.C Birth sort algorithms

Algorithm.3.1.C .

```
void sorted_birth(employee information[],int a){
employee sort[1000];
employee temp;
int i,j;
for (i=0;i<a;i++){

sort[i]=information[i];
}
for(i=0;i<a-1;i++){
for (j = 0; j < a - i - 1; j++) {
if (check_bigger(sort[j].birth.day, sort[j].birth.month, sort[j].birth.year,sort[j +
1].birth.day, sort[j + 1].birth.month, sort[j + 1].birth.year) == 1)
{
temp = sort[j];
sort[j] = sort[j + 1];
sort[j + 1] = temp;
}
}
}
for (i=0;i<a;i++){
printf("%d, %s, %d, %2d-%2d-%4d, %s, 0%ld, %d-%d-%d, %s\n",sort[i].id,
sort[i].name,
sort[i].salary, sort[i].birth.day,
sort[i].birth.month, sort[i].birth.year,
sort[i].address, sort[i].mobile,
sort[i].enrollment.day,
sort[i].enrollment.month,
sort[i].enrollment.year, sort[i].email);
}
}
```

- **Note "check_bigger" function is used to compare between two given date**
- **Its return 0 if the first is bigger and 1 if second is bigger**

CHAPTER 3. Algorithms.

2. search algorithm

The search algorithm take a text from the user and then calculate the number of char using the function **strlen() and then its check if (text[j]==Name [j]) and it's added 1 to count if they matches,**
Then it's compare the count with the chars number in the text.

Algorithm.3.1.A .

```
void search_employee(int a,employee information[]){

char temp[151];
int i,j,count=0,enter=0;
getchar();
printf("Enter employee name: ");
scanf("%150[^\n]",temp);

int n = strlen(temp);

for (i=0;i<a;i++){
count =0;
for (j=0;j<n;j++){
if (information[i].name[j]==temp[j])
count ++;
}

if (count == n){
printf("%d, %s, %d, %d-%d-%d, %s, 0%ld, %d-%d-%d, %s\n",information[i].id,
information[i].name,
information[i].salary, information[i].birth.day,
information[i].birth.month, information[i].birth.year,
information[i].address, information[i].mobile,
information[i].enrollment.day,
information[i].enrollment.month,
information[i].enrollment.year, information[i].email);
enter=1;
}
}
if (enter==0){
printf("No matches found \n");
}
}
```

THANK YOU.